

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

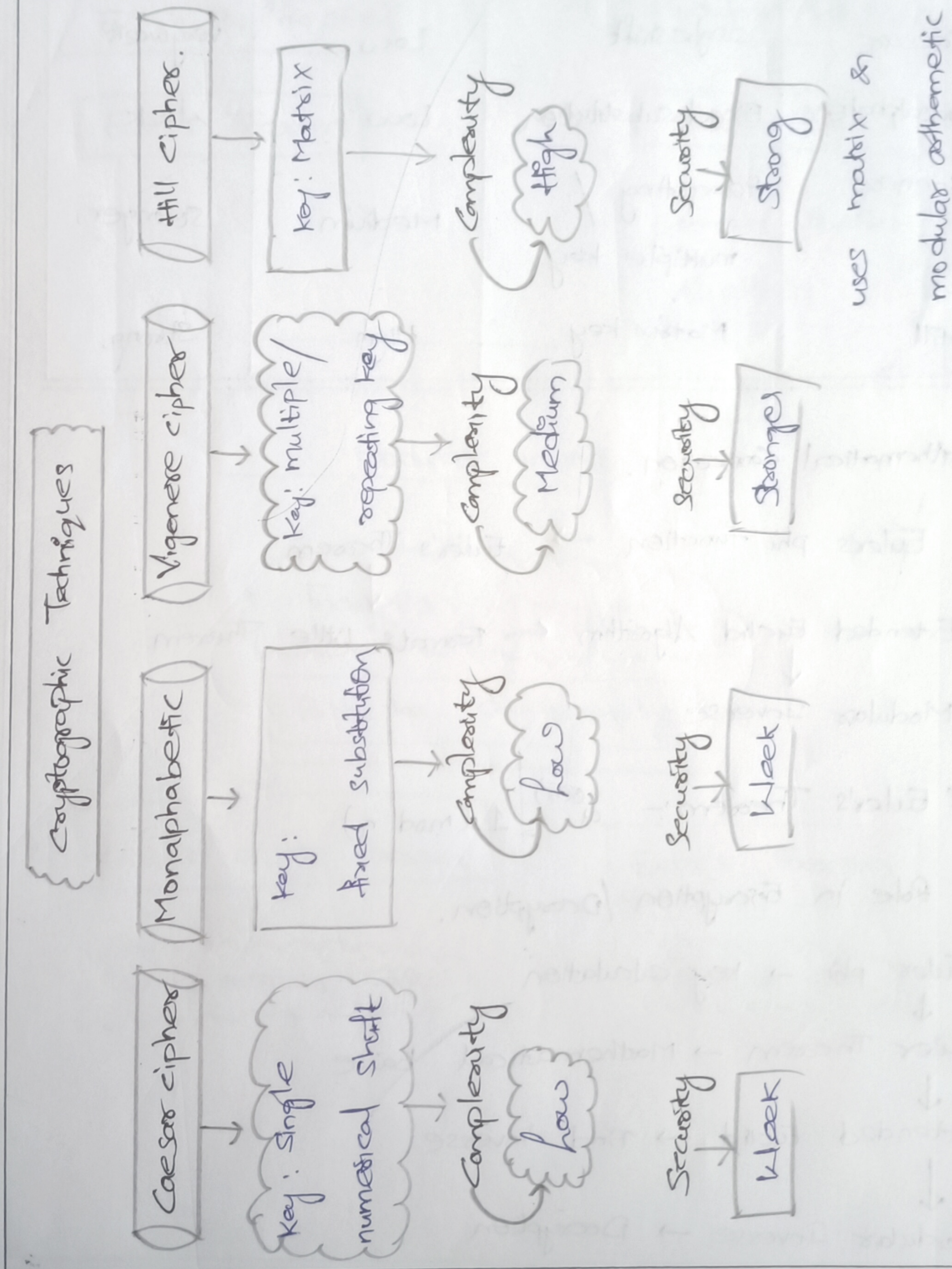
Code of Conduct:

I G. Vignesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

G. Vignesh

Signature of the student:

1. Concept Map: Substitution Techniques in Cryptography.



⇒ Comparison.

Technique	key usage	Complexity	Security
Caesar	single shift	Low	very weak
Monalphabetic	Fixed substitution	Low	Weak
Vigenere	Repeating / multiple key	Medium	Stronger
Hill	Matrix key	High	Strong.

2. i) Mathematical Connection.

Euler's phi function → Euler's Theorem

Extended Euclid Algorithm ← Fermat's Little Theorem

↓
Modular Inverse

* Euler's Theorem:- $a^{\phi(n)} \equiv 1 \pmod{n}$

ii) Role in Encryption / Decryption.

Euler phi → key calculation

↓

Euler Theorem → Mathematical base

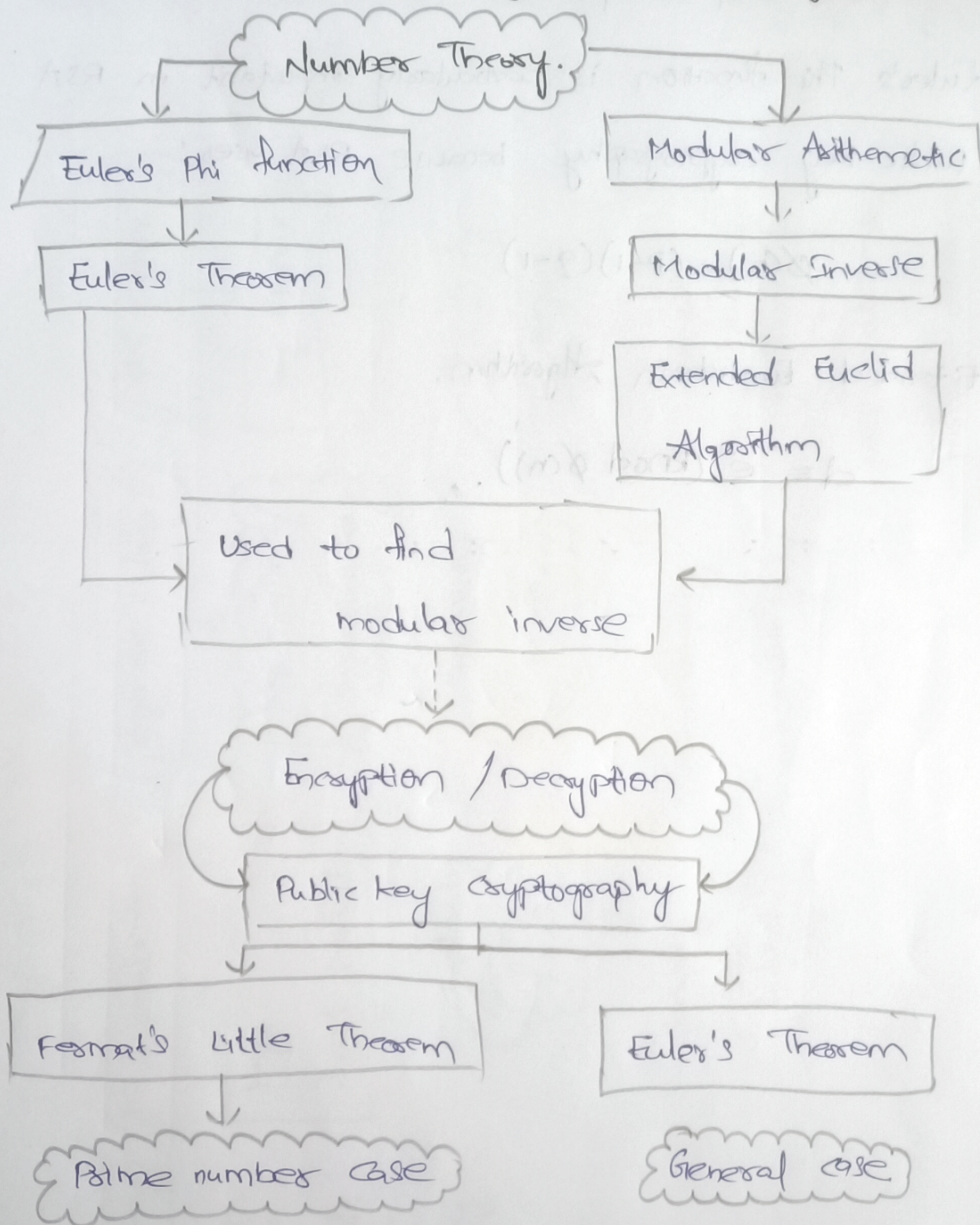
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Extended Euclid → Find inverse

↓

Modular Inverse → Decryption

Concept Map: Number Theory in Cryptography.



iii) Most Critical Concept:

Euler's Phi function is particularly important in RSA public-key cryptography because RSA uses:

$$\phi(n) = (p-1)(q-1)$$

Extended Euclidean Algorithm.

$$d \equiv e^{-1} \pmod{\phi(n)} \dots$$